

Golang Learning Roadmap (9-Month Plan)

Golang Learning Roadmap: Beginner to Master with Interview Preparation (9-Month Plan)

Timeline Summary:

- Months 1–2: Beginner Level
- Months 3–4: Intermediate Level
- Months 5–6: Advanced Level
- Months 7–8: Production-Level Projects + Real-World Practices
- Month 9: Deep Interview Preparation + Final Polishing

Month-by-Month & Weekly Calendar (30 min/day weekdays, 2 hrs/day weekends)

Month 1: Go Basics & Fundamentals

- Week 1: Install Go, setup tools, Hello World, variables, constants.
- Week 2: Functions, if/else, switch, loops.
- Week 3: Arrays, slices, maps.
- Week 4: Structs, methods, simple calculator CLI app.

Month 2: Control, Errors, and Small Projects

- Week 1: Error handling, defer, panic, recover.
- Week 2: Modular code, packages, basic CLI projects.
- Week 3: Word frequency counter.
- Week 4: Build Todo CLI app.

Month 3: Pointers, Interfaces, Concurrency Intro

- Week 1: Pointers, value vs reference.
- Week 2: Interfaces, type assertions.
- Week 3: Project structure, visibility, basic REST API.
- Week 4: Goroutines intro, basic channels.

Month 4: Concurrency & File Handling

- Week 1: Buffered/unbuffered channels, select.
- Week 2: File I/O and CLI file counter project.
- Week 3: Table-driven testing, testing pkg.
- Week 4: Concurrent downloader project.

Month 5: Patterns & Profiling

- Week 1: Worker pool, fan-in/out patterns.
- Week 2: Context handling (timeout, cancel).
- Week 3: Custom errors, middleware basics.
- Week 4: Build rate limiter app.

Month 6: Building REST APIs

- Week 1: RESTful book store (CRUD).
- Week 2: Retry/backoff mechanism.
- Week 3: Profiling using pprof.
- Week 4: Polish all code and push to GitHub.

Month 7: Production Practices

- Week 1: Clean Architecture in Go.
- Week 2: gRPC & Protocol Buffers.

- Week 3: PostgreSQL + sqlx/GORM.
- Week 4: Redis intro, caching mechanism.

Month 8: DevOps + Deployment

- Week 1: Dockerizing Go Apps.
- Week 2: CI/CD using GitHub Actions.
- Week 3: Logging & monitoring (logrus + Prometheus).
- Week 4: Deploy microservice on AWS.

Month 9: Final Projects + Interview Prep

- Week 1: Review Go internals (slices, defer, GC, goroutines).
- Week 2: 25 LeetCode DSA problems in Go (easy + medium).
- Week 3: System design (URL shortener, rate limiter).
- Week 4: Mock interviews, resume polish, final project demo.

Final Projects:

1. Full-featured SaaS backend (Auth, PostgreSQL, Redis, Docker).
2. Real-time messaging/chat app (gRPC or WebSocket).

Study Resources:

- Book: The Go Programming Language by Alan Donovan.
- Video: JustForFunc YouTube.
- Practice: Exercism.io, LeetCode (Golang), Gophercises.
- Advanced: Go Patterns, Learn Go with Tests (quii.gitbook.io).

Stick to the calendar, and you'll be ready to ace any Go interview by the end of 9 months!